Pranav Prasanna Rao

raopranav09@gmail.com | 971-371-8478 | /in/pranavrao09 | pranavrao-dev.vercel.app | github.com/OnMason

EDUCATION

University of California - Irvine **Expected Graduation: June 2028**

Bachelor of Science - Electrical Engineering: *Semiconductors & Optoelectronics Specialization* **GPA: 3.902**

Coursework: Electronics, VLSI Design, Digital Signal Processing, Circuit Design, Semiconductor Devices, Embedded Systems, Computer Architecture, Control Systems, Digital Logic Design, Discrete-Time Signals & Systems, Organization of Digital Computers

EXPERIENCE

Tesla **Sep 2026 - Present**

Hardware Engineering Intern *Palo Alto, CA*

• Automating board-level **DFT** and test development for Keysight i3070 ICT and x1149 **JTAG boundary-scan** platforms, scripting in Python and Korn shell and using **BSDL** device models to build scan chains, sequence execution, and parse parametric measurements

• Engineered custom PCBAs and modular bed-of-nails test fixtures alongside design EEs from schematic capture through assembly and bring-up, applying board-level DFT to route **120+** probe points and reach **94%** nodal access for structural ICT/JTAG coverage

• Executed root-cause failure analysis using oscilloscope profiling, X-ray, and microscopy, tracing intermittent power rail shorts to flexure-induced **MLCC cracking** during fixture actuation, driving a fixture redesign that eliminated board flex and recurring shorts

Revvo Technologies **May 2026 - Aug 2026**

Hardware Engineering Intern *San Mateo, CA*

• Designed a parameterized multi-channel sensor aggregation block in SystemVerilog on an FPGA gateway, implementing a pipelined datapath with FSM-based control to serialize telemetry from BLE valve and tire sensor nodes onto the vehicle bus at **100 MHz**

• Drove RTL-to-bitstream flow in Vivado, closing timing with positive setup slack at **100 MHz** and iterating on synthesis constraints, floorplanning, and implementation strategy to cut dynamic power **15%** and LUT/FF utilization **12%** vs. baseline

• Engineered a Linux-based automated test framework in Python interfacing with a digital oscilloscope, JLink, and programmable power supply unit over GPIB/USB, streaming real-time sensor telemetry and reducing manual bench-characterization time by **40%**

HERO Lab **Mar 2026 - Sep 2026**

Hardware R&D Engineer *Irvine, CA*

• Designed the hardware for a grant-funded contactless heart-sound sensing system, developing a multi-channel MEMS microphone array analog front-end (spanning preamplification, filtering, and ADC conversion) for acquisition of weak acoustic cardiac signals

• Developed a multi-microphone sensor-fusion algorithm that weights each microphone by its SNR and distance to the source to reconstruct clean heart-rate signals, prototyping the pipeline in Python before porting it to embedded firmware for edge execution

• Optimized deep-learning models such as PhysNet and DeepPhys for edge deployment through ONNX quantization, benchmarking model size, transmission latency, and computational efficiency to structure on-device, hybrid, and cloud inference architectures

PROJECTS

5-Stage Pipelined RV32I Processor | *Verilog, Vivado, RTL Design, RISC-V*

• Designed a **5-stage** pipelined **RISC-V** (RV32I) processor in SystemVerilog supporting all **37** user-level instructions, with EX/MEM and MEM/WB forwarding, load-use stall logic, and EX-stage branch resolution with a 2-cycle flush, averaging **1.32** CPI in simulation

• Built a self-checking testbench with constrained-random program generation and a scoreboard against an untimed ISA reference model, running **500** programs (~**218,000** instructions) with **0** mismatches and passing the full **40**-test riscv-tests rv32ui suite

• Wrote **35 SVA** assertions and closed all **117** functional coverage bins, catching **25 of 25** injected RTL bugs during mutation testing

Systolic Array Matrix-Multiply Accelerator | *SystemVerilog, SVA, CDC, RTL Design*

• Designed a parameterized **N×N** output-stationary systolic array for **INT8** matrix multiplication in SystemVerilog, with N^2 multiply-accumulate cells, 32-bit accumulators, and a requantization stage (round, shift, saturate, ReLU) completing in 3N−2 steps

• Built dual-clock asynchronous FIFOs with Gray-coded pointers and two-flop synchronizers for safe clock-domain crossing (**CDC**)

• Verified with constrained-random testbenches and **30 SVA** assertions, running **900** FIFO tests and **4,198** matrix multiplies across four array sizes and five random seeds with **0** mismatches, and catching **22 of 24** deliberately injected bugs in mutation testing

Optical Material Classifier | *Photodiode Sensing, Mixed-Signal Circuit, ESP32*

• Designed an RGB LED light-refraction mixed-signal circuit with photodiode sensing across **2+** functional prototypes, differentiating plastic types (**HDPE**, **PET**) by their wavelength-dependent reflectance, without external lab equipment or user input

• Built the analog capture chain feeding ESP32 processing, converting raw photodiode response into real-time material classification

SKILLS

Hardware: SystemVerilog, Verilog/RTL, RISC-V, FPGA, Functional Verification, PCB Design, Schematic Capture, I2C, UART, SPI, CAN
Software: Vivado, Cadence, Altium, LTSpice, KiCad, Python, C, C++, Assembly, Tcl, Make, MATLAB, GDB, Linux, Git
AI/ML: PyTorch, ONNX, MediaPipe, NumPy, OpenCV
Interests: VLSI, ASIC, Embedded Systems, Semiconductor Design, Power Systems, AI agents, Badminton, Gym, Anime